Soumyadip Kundu

CURRICULUM VITAE

Contact

Email: soumyadip23@iisertvm.ac.in personal website

Education

Ph.D. in Physics Jul 2023 – Present
IISER Thiruvananthapuram, India
Supervisor: Dr. Tanumoy Mandal

M.Sc. in Physics Aug 2021 – Jun 2023
IISER Thiruvananthapuram, India

B.Sc. in Physics Aug 2018 – Jul 2021
Jadavpur University, Kolkata, India

Research Interests

Beyond-Standard-Model (BSM) Collider phenomenology.
Effective Field Theories
Precission studies.

Publications

Preprints

1. U. Danielsson, R. Enberg, G. Ingelman, **S. Kundu**, T. Mandal, and S. Mitra, "Multi-Higgs production from a theory of varying Higgs quartic and top Yukawa couplings," arXiv:2610.21751 [hep-ph].
2. U. Danielsson, R. Enberg, G. Ingelman, **S. Kundu**, T. Mandal, and S. Mitra, "Testing varying coupling constants through multi-Higgs production at the LHC," arXiv:2606.14303.

Journal Articles

1. A. Bhaskar, D. Das, **S. Kundu**, A. A. Madathil, T. Mandal, and S. Mitra, "Vector leptoquark contributions to lepton dipole moments," *Phys. Rev. D* **111**, 015045 (2025), arXiv:2408.11798 [hep-ph].

Conference Proceedings

1. T. Mandal, A. Bhaskar, A. Das, **S. Kundu**, S. Mitra, and R. Sharma, "Exclusion limits and unconventional searches for leptoquarks," *Springer Proc. Phys.* **322**, 185–189 (2026).
2. **S. Kundu**, A. Bhaskar, D. Das, A. A. Madathil, T. Mandal, and S. Mitra, "Exploring TeV scale vector leptoquark as a solution to lepton dipole moments," (2026).

Teaching Experience

Teaching Assistant–IISER Thiruvananthapuram

PHY/MSP 314: Quantum Mechanics I (Varsha 2023,24; Vasanth 25)

PHY/MSP 316: Statistical Mechanics I (Vasanth 2024)

PHY/MSP 327: Quantum Mechanics II (Varsha 2025)

Minor project mentoring–IISER Thiruvananthapuram (Varsha 2023)
Supervised an M.Sc. first-year student on "Statistical Analysis of Higgs Boson Discovery: Evaluating Significance through the p-value".

Talks and Posters

Poster: Exploring TeV-scale vector leptoquark as a solution to lepton dipole moments Dec 2024
XXVI DAE-BRNS High Energy Physics Symposium, BHU, Varanasi

Poster: Leptoquarks and lepton dipole moments Feb 2025
Frontier Symposium in Physics (FS-PHY 2025)

Conferences and Workshops

Advanced School on Mathematical Methods for Physicists — 2026
IMSc, Chennai

SSHEP 2025: Spring School on High Energy Physics — 2025
IMSc, Chennai

GIAN Course: Standard Model Effective Field Theories and Applications to Higgs, Neutrinos, and Dark Matter — 2024
IIT Guwahati

DAE-HEP 2024: XXVI DAE-BRNS High Energy Physics Symposium — 2024
BHU, Varanasi

PPC 2024: 17th International Conference on Interconnections between Particle Physics and Cosmology — 2024
IIT Hyderabad

ML4HEP 2023: Statistical Methods and Machine Learning in High Energy Physics — 2023
ICTS, Bengaluru

Awards and Honors

MSc with Distinction (CGPA: 9.0/10) — 2023

Qualified GATE (Physics) — 2023

Qualified JAM (Physics) — 2021

DST-INSPIRE Scholarship — 2018 – 2023

Skills

Programming: Python, Mathematica, Package-X, MadGraph
Relevant coursework: Quantum Field Theory I, Particle Physics, General Relativity, Quantum Information Theory.
Languages: English, Bengali, Hindi

Extra-Curricular Activities

Contributor, PSIt
IISER TVM Physics Society journal
Wrote physics articles in Bengali for science outreach.

Boys Hostel Representative (Mukurthi)
IISER Thiruvananthapuram
Over two years: liaison between students and administration, and organizer of hostel activities.

References

Dr. Tanumoy Mandal, Assistant Professor, IISER Thiruvananthapuram
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Dr. Subhadip Mitra, Assistant Professor, IIIT Hyderabad
subhadip.mitra@iiit.ac.in

Dr. Shabnam Iyyani, Assistant Professor, IISER Thiruvananthapuram
shabnam@iisertvm.ac.in